AQUAFAB INDUSTRIES

Swim Performance (Ucrit) Testing

The stepped-velocity test, with optional tracking and cost of transport.

For Aquafab Workbench 0.2.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Swim performance (U_{crit}) testing

The critical swimming speed test of Brett (1964): water speed rises in fixed steps at fixed intervals until the animal can no longer hold station. You press **Fish Fatigued** at that moment, and U_{crit} is interpolated:

> $U_{crit} = U + (T / T_i) \times U_i$

U is the last speed held for a full interval, U_i the step size, T_i the interval and T the time at the final speed.

You need: a machine with a variable-speed pump (the **Swim Performance** tab appears only then), and a license to start a test on connected hardware. A camera is optional.

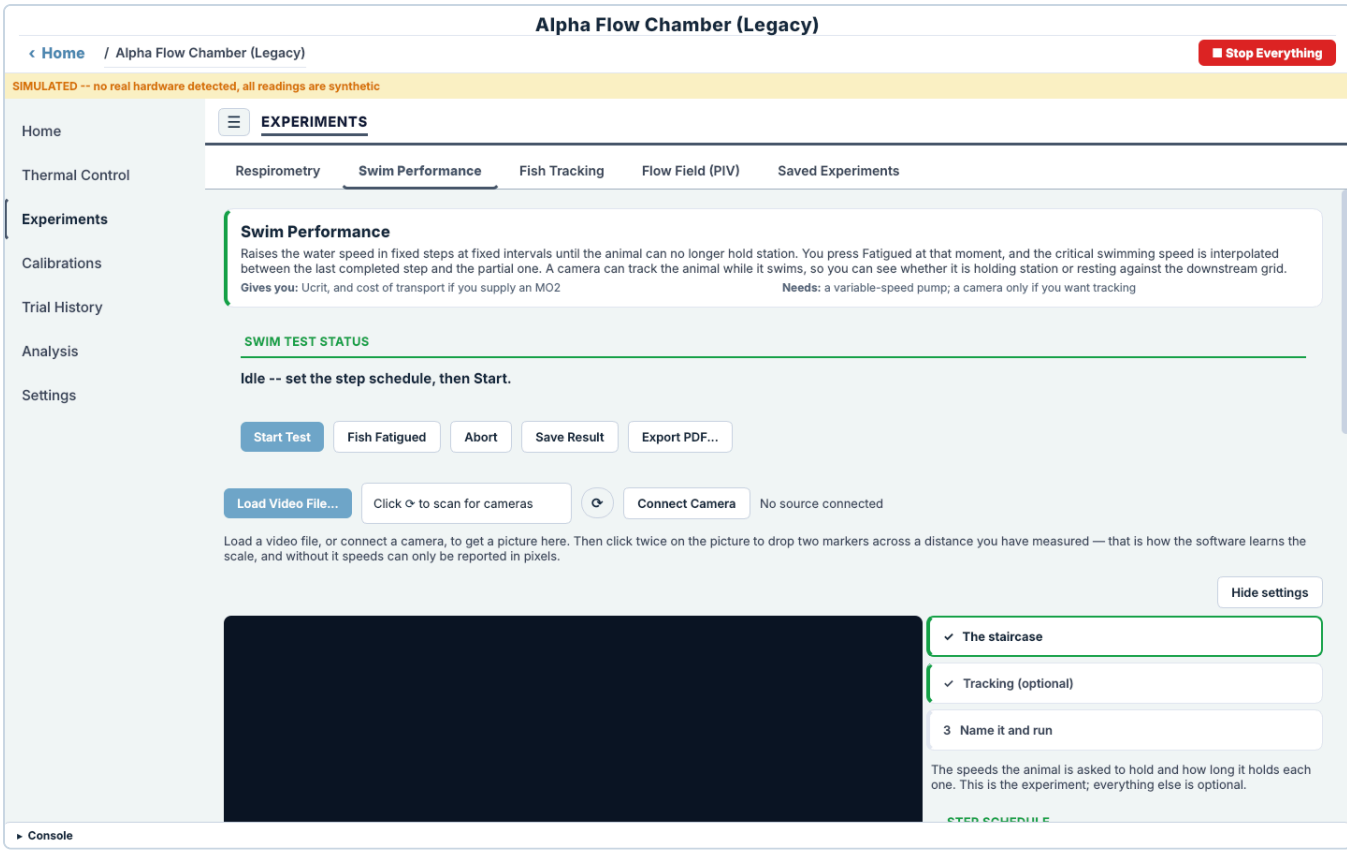
> Screenshots are from a simulated machine. Numbers in them are not measurements.

1.1 Before the day: calibrate velocity

The motor is commanded in per cent of output. To report U_{crit} in cm/s, the software needs to know how many cm/s one per cent produces. Measure it with the **Chamber Velocity (PIV)** calibration (*PIV Flow Measurement and Chamber Velocity Calibration*, Part 3). Without it, U_{crit} is reported in % output, which cannot be compared between machines.

1.2 Set up

Experiments → Swim Performance.



Swim Performance.

1.2.1 The staircase

Field	What to enter
Motor	The pump that drives the working section.
Start output (%)	The first speed.
Step size (%)	How much each step adds.
Interval T_i (s)	How long each speed is held (default 1200 s; published protocols range from minutes to an hour).
Maximum output (%)	The staircase never goes above this.
cm/s per % output	From the velocity calibration. 0 keeps results in %.
Fish length (cm)	For U_{crit} in body lengths per second. 0 skips it.
MO_2 at speed (mg/kg/h)	Only if you measure oxygen alongside: gives cost of transport at each step and U_{opt}, the cheapest speed. 0 skips it.
Simulation speed (x)	Simulated machines only: rehearse a long test in minutes.

1.2.2 Tracking (optional)

A track of the animal is the evidence for the moment you called fatigue.

1. **Connect Camera** above the picture.
2. With the tunnel **empty**, press **Capture background**.
3. Set the scale: click two points a known distance apart, type the distance, **Apply scale from markers**.
4. Tick **Track the animal during the test**.

The detection settings are explained in the *Fish Tracking* guide.

1.2.3 Name it and run

Name, **Treatment** and **Notes**. **Save experiment** keeps the staircase for the next animal.

1.3 Run

1. Put the animal in and let it settle at a low speed, as your protocol requires.
2. Press **Start Test**. The status shows the current speed and the time left in the step.
3. Watch the **animal**, not the screen. Decide your fatigue criterion before the test (typically: resting against the downstream grid and not resuming after gentle encouragement) and apply it the same way every time.
4. The moment it fatigues, press **Fish Fatigued**, or the **F** key. The motor stops.

Abort stops the test and the motor without a result. If the staircase reaches **Maximum output**, it holds there and keeps timing until you press Fish Fatigued or Abort.

1.4 The result

The **Result** card shows U_{crit} in cm/s (if calibrated), in body lengths per second (if a length was given) and in % output; the completed steps and the time at the final speed; and, with an MO_2, the cost of transport at each step and U_{opt}.

Press **Save Result** to keep it in Trial History. **It is not saved until you do. Export PDF…** prints the protocol, result and speed chart (needs a license).

1.5 Good practice

- A large animal in a small working section swims in faster water than the nominal speed. The Fish Tracking page has a solid-blocking correction (Bell & Terhune 1970); verify its coefficients before publishing.
- Keep the interval and step size the same within a study. U_{crit} depends on both.